

Daratumumab, lenalidomide, dexamethasone (DRd)

DRd is a cancer drug treatment. It is used to treat a type of blood cancer called myeloma.

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What is DRd?

DRd is a combination of cancer drugs. It is used to treat a type of blood cancer called [myeloma](#).

DRd includes the following drugs:

- daratumumab (Darzalex[®]) – a type of [targeted therapy](#) drug called a monoclonal antibody
- lenalidomide (sometimes called by a brand name Revlimid[®]) – a type of targeted therapy and also an immunotherapy drug.
- dexamethasone – a [steroid](#).

It is best to read this information with our general information about [cancer drug treatment](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How DRd is given

You usually have daratumumab at a chemotherapy day unit or clinic as an outpatient. Lenalidomide and dexamethasone are capsules or tablets that you take at home.

During your course of treatment, you will meet someone from your cancer team, such as a:

- blood cancer doctor (haematologist)
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of each treatment, you will have a blood test. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your course of treatment

You usually have a course of several cycles of DRd over a few months. Each cycle lasts 28 days (4 weeks).

You will have daratumumab once every week for the first 2 cycles. Then once every 2 weeks for the next 4 cycles. Then every 4 weeks.

You take lenalidomide once a day on day 1 to 21 of each 4 week cycle.

You take dexamethasone once or twice a week.

You usually keep having this treatment as long as it is effective and any side effects can be managed. Your cancer team will discuss your treatment plan with you. They may give you a copy of the plan to take home.

Having daratumumab

Daratumumab can be given as a drip into a vein (intravenous). But for this combination, you usually have daratumumab as an injection under the skin (subcutaneous). Your nurse will give the injection into your tummy area (abdomen). They will give it over 3 to 5 minutes. The nurse will give it in different areas of the tummy each time. This can help stop your skin from becoming sore.

You will have medicines before and after each injection to reduce the chance of an allergic reaction. After the first treatment, you may be asked to wait in the hospital for 4 to 6 hours after the injection. This is to check for any signs of a reaction to the treatment.

Taking lenalidomide and dexamethasone

These drugs come in capsules or tablets. This means you can take them at home.

Your doctor, nurse or pharmacist will explain when and how to take each drug. You may be given capsules or tablets of different strengths. Make sure you understand which days you should take each drug and what dose you should take.

They may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they have been explained to you.

Swallow the lenalidomide and dexamethasone whole with a glass of water. Do not chew, open, break or crush the capsules or tablets.

You can take lenalidomide with or without food. This drug can make you sleepy so you may want to take it in the evening.

You should take dexamethasone after food, or with plenty of water. This drug can cause difficulty sleeping so it is best to take with breakfast or lunch.

If you forget to take dexamethasone, take the missed dose as soon as possible.

If you forget to take lenalidomide and there is less than 12 hours until your next dose, do not take the missed dose. Take the next dose at the usual time and tell your doctor, nurse or pharmacist. Do not take a double dose.

Other things to remember about your capsules or tablets:

- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.

Preventing pregnancy while taking lenalidomide

You must not [become pregnant or make someone pregnant](#) while taking lenalidomide. This is because it may cause severe abnormalities in developing babies.

You will take part in a Pregnancy Prevention Programme during treatment. Even if you do not think you are at risk of getting pregnant or making someone pregnant, you will be included in this programme.

Your doctor, nurse or pharmacist will give you information about the programme and the risks of lenalidomide and pregnancy. They will ask you to sign a [consent form](#) to show you understand the information.

Pregnancy tests and contraception

If you are of childbearing age and able to get pregnant, you will have a pregnancy test before starting this treatment. The pregnancy test is repeated every 4 weeks during treatment, and 4 weeks after treatment finishes.

You must use an effective form of contraception, such as an implant, injection or the progesterone-only pill. You should not use the combined oral contraceptive pill. It increases your risk of developing blood clots.

You must use contraception to prevent pregnancy:

- for 4 weeks before treatment
- during treatment
- for 4 weeks after treatment finishes.

If you think you may be pregnant at any time during your treatment, contact your doctor, nurse or pharmacist straight away.

Contraception to prevent making someone pregnant

Lenalidomide can pass into the semen that carries sperm. During treatment and for a week after treatment finishes, you must not:

- make someone pregnant
- donate sperm.

You can protect partners by using a condom during sex. During treatment and for a week after treatment finishes, you must use a condom during sex with anyone who is:

- pregnant

- of childbearing age and able to get pregnant.

If a partner thinks they might be pregnant during your treatment, contact your doctor, nurse or pharmacist straight away.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having this treatment or shortly after they have it:

Allergic reaction

Some people have an allergic reaction while having this treatment. The first treatment of daratumumab is the most likely to cause a reaction. Before the injections, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching

- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- a blocked or tickly nose
- a cough or tickly throat
- swelling of your face or mouth
- pain in your back, tummy or chest
- blurred vision.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number. Always call 999 if swelling happens suddenly or you are struggling to breathe.

Problems at the injection site

If you are having daratumumab as an injection in the tummy, the skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

Tell your doctor, nurse or pharmacist if this happens. They will give the injection in a different areas each time.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Feeling sick

You may feel sick. This is usually mild, but your doctor, nurse or pharmacist can prescribe anti-sickness drugs to help. Tell them if you still feel sick or it gets worse. They can give you advice or a different anti-sickness drug to try.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Mood changes and behaviour changes

This treatment can affect your mood and behaviour. It can cause:

- feelings of anxiety or restlessness
- mood swings (moods that go up and down)
- low mood or depression.

Sometimes, when taken in higher doses, dexamethasone can cause confusion or changes in thinking. This can include having strange or frightening thoughts.

If you notice any changes in your mood or behaviour, tell your doctor, nurse or pharmacist. They may make some changes to your treatment if the side effects are causing you problems.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital.

Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Muscle spasms and pain

This treatment can cause muscle spasms. They are usually mild. Or you may get pain in your muscles or joints. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Changes in blood pressure

This treatment may affect your blood pressure. Tell your nurse, doctor or pharmacist if you have headaches or feel dizzy. These can be signs of blood pressure changes.

Effects on eyesight

This treatment can affect your eyesight. Tell your doctor, nurse or pharmacist if you notice any change in your vision.

Tummy pain or indigestion

Dexamethasone can irritate the stomach lining. If you have indigestion or pain in your tummy, tell your doctor, nurse or pharmacist. They can prescribe drugs to help reduce stomach irritation.

To help protect your stomach, you should take the dexamethasone with food.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Blood clot risk

This treatment increases your risk of a blood clot. Your cancer team will give you drugs to take that help prevent blood clots.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
-

reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker

- suddenly feeling breathless or coughing.

Always call 999 if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Hearing changes

This treatment may cause hearing changes, including hearing loss. You may have ringing in the ears. This is called tinnitus. You may also become unable to hear some high-pitched sounds. Hearing changes may get better after this treatment ends. But this does not always happen. If you notice any changes in your hearing, tell your doctor, nurse or pharmacist.

Difficulty getting an erection

You may have difficulty getting or keeping an erection while on this treatment.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

- a headache that does not get better
- drowsiness or confusion
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Second cancer

This treatment may increase your risk of developing a different type of cancer. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can talk to you about this.

Other important information

Blood transfusions

Daratumumab can affect the result of a blood test called the indirect Coombs test. This is one of the tests that matches you to a donor's blood before you have a blood transfusion.

Before you start daratumumab, you will have the blood tests you need for a blood transfusion. Your doctor may also give you a card to carry that explains the treatment you are having.

If you need a blood transfusion during treatment or for 6 months after, tell the person arranging it that you have had daratumumab. You can show them the card your doctor has given you.

Hepatitis B and shingles

Some viral infections can become active again while you are taking this treatment. This includes hepatitis B (a liver infection) and herpes zoster (shingles). These may need treatment with anti-viral drugs. Your doctor or nurse will talk to you about this. Before you start treatment, you will have a blood test to check if you have had hepatitis B.

Driving

This treatment may affect your ability to drive. Do not drive if you have blurred vision, or if you feel dizzy, very tired or sleepy. Talk to your doctor, nurse or pharmacist if you are not sure whether it is safe for you to drive.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch



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